Seungjae Han

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stevejayh.github.io

RESEARCH INTERESTS

AI-Driven Microscopy & Computational Neuroscience: How can we analyze brain activity data at scale?

I develop computational and AI-driven tools for neuroscience, spanning algorithm design to imaging pipelines. My work spans neuroscience, electrical engineering, and computer vision, with the broader goal of enabling new discoveries in brain science through principled algorithmic design and data-efficient learning.

(Computational Neuroscience · Machine Learning · Biomedical Imaging)

EMPLOYMENT

2026- **KAIST** Advisor : Young-Gyu Yoon
Postdoctoral Fellow (Sejong Science Fellow; Jang Youngsil Fellow)

EDUCATION

2020-26 **KAIST** Advisor : Young-Gyu Yoon
Ph.D. Candidate in School of Electrical Engineering

2017-20 **Yonsei University** Advisor : Jiwon Seo
Bachelor of Science in School of Integrated Technology

SELECTED PUBLICATIONS

** co-first authors, ** co-corresponding authors*

Self-supervised video processing with self-calibration on an analogue selector-less memristor array-based platform
H. Jeong* , **S. Han***, S.-O. Park, T. R. Kim, J. Bae, T.-H. Jang, Y. Cho, S. Seo, H.-J. Jeong, S. Park, T. Park, J. Oh, J. Park, D. Jeon, I. Kwon , Y.-G. Yoon**, S. Choi**
Nature Electronics, Jan. 2025. [🏆 **Introduced in Nature Electronics News & Views, Nature Electronics Editorial, KAIST Breakthroughs**]

Statistically unbiased prediction enables accurate denoising of voltage imaging data
M. Eom*, **S. Han***, P. Park*, G. Kim, E.-S. Cho, J. Sim, K.-H. Lee, S. Kim, H. Tian, U. L. Böhm, E. Lowet, H. Tseng, J. Choi, S. E. Lucia, S. H. Ryu, M. Rózsa, S. Chang, P. Kim, X. Han, K. D. Piatkevich, M. Choi, C.-H. Kim, A. E. Cohen, J.-B. Chang, Y.-G. Yoon
Nature Methods, Sep. 2023. [🏆 **Selected as the Cover and KAIST Breakthroughs**]

SELECTED AWARDS AND HONORS

2026	**Outstanding Ph.D. Dissertation Award**	KAIST EE
	Jang Young Sil Fellow (Excellence Track)	KAIST
	Sejong Science Fellowship	National Research Foundation of Korea
2024	**Graduate Student Outstanding Paper Award**	KAIST
	Top reviewer of NeurIPS 2024 & Complimentary registration	NeurIPS
2023	**Trainee Professional Development Award**	Society for Neuroscience
	AKN Outstanding Research Award (IBS/AKN Pre-doctoral award)	Association of Korean Neuroscientists
	Outstanding Teaching Assistant Award (Course: Signals and Systems)	KAIST EE
2020-2025	**Government-sponsored scholar**	KAIST
2019	**Undergraduate Research Program (Research fund) & Outstanding Project Award**	Korea Foundation for the Advancement of Science & Creativity
2017-19	**IT Consilience Creative Program (Tuition waiver & stipend)**	Ministry of Science, ICT and Future Planning

PUBLICATIONS

** co-first authors, ** co-corresponding authors*

2026 **High speed imaging system**
J. Yoon*, S. Kim*, **S. Han***, ... , Y.-G. Yoon**, M. Choi**
PhotoniX (in revision)

2025 **Nanoscale resolution imaging of the whole mouse embryos and larval zebrafish using expansion microscopy**
J. Sim*, C. E Park*, I. Cho*, K. Min, M. Eom, **S. Han**, H. Jeon, E.-S. Cho, Y. Lee, Y. H. Yun, S. Lee, D.-H. Cheon, J. Kim, M. Kim, H.-J. Cho, J.-W. Park, A. Kumar, Y. Chong, J. S. Kang, K. D. Piatkevich, E. E. Jung, D.-S. Kang, S.-K. Kwon, J. Kim, K.-J. Yoon, J.-S. Lee, C.-H. Kim, M. Choi, J. W. Kim, M.-R. Song, H. J. Choi, E. S. Boyden, Y.-G. Yoon**, J.-B. Chang**
ACS Nano

Design Principles of Multi-Scale J-invariant Networks for Self-Supervised Image Denoising
H. Yu*, **S. Han***, Y.-G. Yoon
*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**)* [🏆 **Oral**]

2024 **In Vivo Optical Clearing of Mammalian Brain**
G. T. Franzesi*, I. Gupta*, M. Hu, K. Piatkevich, M. Yildirim, J.-P. Zhao, M. Eom, **S. Han**, D. Park, H. Andaraarachchi, Z. Li, J. Greenhagen, A. M. Islam, P. Vashishtha, Z. Yaqoob, N. Pak, A. D Wissner-Gross, D. A. Martin-Alarcon, J. J. Veinot, P. T. C. So, U. Kortshagen, Y.-G. Yoon, M. Sur**, E. S. Boyden**
bioRxiv

From Pixels to Information: Artificial Intelligence in Fluorescence Microscopy
S. Han, J. Y. You, M. Eom, S. Ahn, E.-S. Cho, Y.-G. Yoon
Advanced Photonics Research

2023 ***In vivo* whole-brain imaging of zebrafish larvae using three-dimensional fluorescence microscopy**
E.-S. Cho, **S. Han**, G. Kim, M. Eom, K.-H. Lee, C.-H. Kim, Y.-G. Yoon
Journal of Visualized Experiments

Robust and Efficient Alignment of Calcium Imaging Data through Simultaneous Low Rank and Sparse Decomposition
J. Cho*, **S. Han***, E.-S. Cho, K. Shin, Y.-G. Yoon
*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**)*

2022 **IMPASTO: Multiplexed cyclic imaging without signal removal via self-supervised neural unmixing**
H. Kim*, S. Bae*, J. Cho, H. Nam, J. Seo, **S. Han**, E. Yi, E. Kim, Y.-G. Yoon**, J.-B. Chang**
bioRxiv

Three-dimensional fluorescence microscopy through virtual refocusing using a recursive light propagation network
C. Shin*, H. Ryu*, E.-S. Cho, **S. Han**, K.-H. Lee, C.-H. Kim, Y.-G. Yoon
Medical Image Analysis

2021 **3DM: Deep decomposition and deconvolution microscopy for rapid neural activity imaging**
E.-S. Cho*, **S. Han***, K.-H. Lee, C.-H. Kim, Y.-G. Yoon
Optics Express [🏆 **Selected as Editor's pick and Image of the Week**]

Efficient Neural Network Approximation of Robust PCA for Automated Analysis of Calcium Imaging Data
S. Han, E.-S. Cho, I. Park, K. Shin, Y.-G. Yoon
*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*

2019 **Smartphone Application to Estimate Distances from LTE Base Stations Based on Received Signal Strength Measurements**
S. Han, T. Kang, J. Seo
*International Technical Conference on Circuits/Systems, Computers and Communications (**ITC-CSCC**)*

2018 **Observation of Human Trajectory in Response to Haptic Feedback from Mobile Robot**
H.-S. Moon, W. Kim, **S. Han**, J. Seo
*International Conference on Control, Automation and Systems **(ICCAS)***

INVITED TALKS

2023 **SUPPORT: Versatile denoising AI for microscopy data**
Korea Institute of Science and Technology (KIST) (2023. 10.)

PROFESSIONAL SERVICE

Reviewer **NeurIPS** 2025 (5 papers) 2024 (6 papers, **Selected as Top reviewer**), 2023 (5 papers)
ICLR 2024 (3 papers), **AAAI** 2025 (4 papers)
MICCAI 2024 (4 papers), 2023 (3 papers), 2022 (5 papers)

MENTORING EXPERIENCE

2025 **Juno Cho** (Undergraduate Student at KAIST)
Image processing, Cell segmentation

2024 **Hayeong Yu** (Undergraduate Student - M.S. student at KAIST)
Self-supervised learning, Image processing, Denoising (paper published on WACV)

2021 **Eunsu Kim** (Undergraduate Student at KAIST - M.S. student at KAIST)
Machine learning basics, Processing multiplexed images (preprint released on bioRxiv)

TEACHING EXPERIENCE

Teaching assistant (TA)

2023 **Machine learning and Big data (Expert course)**, Seongnam-KAIST Center For Next Generation ICT
Course for general public and office workers, about Reinforcement Learning, **Head TA**

Signals and Systems (EE205), KAIST
Introductory level course, **Head TA**, [🏆 **Best Teaching Assistant Award**]

2022 **Electronics Design Lab (EE305)**, KAIST
Undergraduate level course, about Circuits

Special Topics in Electrical Engineering <AI Capston Design> (EE488), KAIST
Senior level course, about Reinforcement Learning, **Head TA**

2021 **Electronics Design Lab (EE405A)**, KAIST
Senior level course, about Robotics

Basics of Artificial Intelligence (CoE202A), KAIST
Introductory level course, about Computer Vision

2020 **Basics of Artificial Intelligence (CoE202A)**, KAIST
Introductory level course, about Computer Vision

REFERENCES

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ygyoon@kaist.ac.kr

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Jiwon Seo
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